

seafar: Sparse Exploratory Approximate Factor Analysis for High-dimensional Data in R

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Abstract

This manuscript provides a tutorial for the package *seafar* in the statistical software R to perform a Sparse Exploratory Approximate Factor Analysis for high-dimensional data. The methods used in this package make use of the approximate factor model together with cardinality constraint. This helps deal with exploratory high-dimensional settings and complex models characterized by many parameters for a limited sample size, in which most latent variable methods fail. The objectives of this package are two-fold: i) provide users with easy control over the exact number of nonzero/zero loadings for the simple structure of the factor loadings; ii) model selection procedure to help inform users about the simple structure in an exploratory setting with little to no prior information; iii) extract factor scores which can be used in further analyses. The use of *seafar* package is demonstrated using two empirical data sets.

Impact Statement

This template helps you get started.

Keywords: exploratory factor analysis, high-dimensional data, regularization, R package

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Studying latent constructs, such as personality, intelligence, and well-being, is at the core of behavioral research. However, it is met with new challenges introduced by the “Big Data” era, characterized by intensive data collection and technologically advanced measurement tools. Specifically, researchers often need to deal with data sets that have more variables than the number of observations, or so-called high-dimensional low-sample-size (HDLSS) data. Furthermore, new types of data (e.g., GPS or physiological data collected using smart watches) are often exploratory in nature with limited or no prior theoretical knowledge, which can lead to complex models with many unknown parameters relative to the number of observations. Thus, existing (exploratory) latent variable methods often fail to work in these settings, yielding unstable and/or improper solutions or having non-convergent issues. We refrain from discussing in detail these methods and their issues in this paper; however, interested readers are referred to (Le et al., 2026) for comparative overviews.

Regularized ESEM (Le et al., 2026) was proposed to solve the abovementioned issues. This two-stage method relies on the approximate factor analysis framework [citation] to extract latent factors underlying high-dimensional data. In the first stage, Regularized ESEM imposes simple structure on the loading matrix through cardinality constraint for orthogonal factors and the LASSO penalty for the non-orthogonal factors. The authors proposed a sequential model selection strategy to tune the algorithm: first, the number of factors is determined, after which the number of nonzero loadings (cardinality value or LASSO penalty parameter) is chosen to maximize the Index of Sparseness (Gajjar et al., 2017; Gu et al., 2019; Trendafilov, 2014). The current manuscript introduces an R package called *seafar* to implement this first stage with a few modifications. Firstly, instead of having two types of regularization (i.e., cardinality constraint and LASSO penalty) separately for orthogonal and non-orthogonal factors, *seafar* implements a unified algorithm that uses cardinality constraint for both orthogonal and non-orthogonal factors. Secondly, *seafar* offers the original model selection procedure with the Index of Sparseness and an alternative procedure using warm starts (Friedman et al., 2010) which can help reduce the

computational time.

The rest of the paper is structured as follows: first, we introduce the Sparse Approximate Factor Analysis (SEAFa) model and its model selection procedures; next, the use of the *seafar* package is demonstrated using two empirical data sets. The paper is concluded with some remarks.

Method

In this section, we will introduce the SEAFa model, its estimation, and model selection procedures.

The SEAFa model

Let $\mathbf{y}$ be a $N \times 1$ vector of standardized items (i.e., items are mean-centered and scaled to unit variance), $\mathbf{P}$ be a $J \times P$ matrix of factor loading matrix, and $\boldsymbol{\eta}$ be a $Q \times 1$ vector of factor scores, we have the SEAFa model as follows:

$$\mathbf{y} = \mathbf{P}\boldsymbol{\eta} + \boldsymbol{\epsilon} \quad (1)$$

where $\boldsymbol{\epsilon}$ is the $J \times 1$ vector of residuals. Model (1) is subject to two identification constraints: 1) $\mathbf{P}$ is assumed to show simple structure (i.e., not all items load on all factors but some/many have zero loadings), and 2) factor scores $\boldsymbol{\eta}$ are subject to length restriction. Furthermore, the residuals $\boldsymbol{\epsilon}$ are assumed to have mean zero and uncorrelated with the factor scores. Here, $\boldsymbol{\epsilon}$ are allowed to be weakly correlated, which is an assumption that we relax from the common factor model. With this assumption, the model

There are several ways to estimate the approximate factor model (e.g., maximum likelihood-based (Bai & Li, 2016), covariance matrix estimation (Fan et al., 2021)). Here, we chose to use a least-squares approach as it has been shown to have good properties when dealing with high-dimensional data (Fan et al., 2021). Let p_{jq} denote the loading of variable j on factor q and η_{iq} denote the score of person i on factor q , our objective function is as follows:

$$\begin{aligned}
& \arg \min_{\eta, p} \sum_{j=1}^J \sum_{i=1}^N (y_{ij} - \sum_q \eta_{iq} p_{jq})^2 \\
& \text{subject to } \sum_i \eta_{iq}^2 = N \quad \forall q = 1, \dots, Q, \\
& \text{and Card}(\mathbf{P}) = C,
\end{aligned} \tag{2}$$

or in matrix form:

$$\begin{aligned}
& \arg \min_{\mathbf{H}, \mathbf{P}} \|\mathbf{Y} - \mathbf{H}\mathbf{P}^T\|_2^2, \\
& \text{subject to } \boldsymbol{\eta}_q^T \boldsymbol{\eta}_q = N, \\
& \text{and Card}(\mathbf{P}) = C,
\end{aligned} \tag{3}$$

with $\text{Card}(\mathbf{P})$ being the number of nonzero elements in the loading matrix $\mathbf{P}$. To solve (3), we use an alternating optimization (AO) scheme: factor scores $\mathbf{H}$ and loadings $\mathbf{P}$ are updated in turn based on fixed values of the other. Specifically, with fixed $\mathbf{P}$, $\hat{\mathbf{H}}$ (subject to norm constraint) is obtained using the method of Lagrange multipliers. $\hat{\mathbf{P}}$ is estimated with fixed $\mathbf{H}$ via the cardinality-constrained linear regression problem:

$$\begin{aligned}
& \hat{\mathbf{P}} = \arg \min_{\mathbf{P}} \|\text{vec}(\mathbf{Y}) - (\mathbf{I} \otimes \mathbf{H})\text{vec}(\mathbf{P}^T)\|_2^2, \\
& \text{subject to Card}(\mathbf{P}) = C,
\end{aligned} \tag{4}$$

where $\otimes$ is the Kronecker product. $\hat{\mathbf{P}}$ is updated using a numerical solution proposed by Adachi and Kiers (2017) (i.e., special case of a MM procedure) as:

$$\text{vec}(\mathbf{P}^u) = T_C(\text{vec}(\mathbf{P}_{old}^T) - \frac{1}{N\alpha}(\mathbf{H}^T \mathbf{H} \mathbf{P}_{old}^T - \mathbf{H}^T \mathbf{Y})), \tag{5}$$

with α being the maximum eigenvalue of $\mathbf{H}^T \mathbf{H}$ and T_C denotes the vector whose C elements with

the smallest absolute values are put to zeros. One can also see this update as a solution to a projected gradient scheme with fixed step size α^{-1} . Since the main complexity here involves sorting the loading matrix $\mathbf{P}$ of size $J \times Q$, the update can efficiently scale up to high-dimensional setting with very large J .

In the special case of orthogonal factors, that is, $\mathbf{H}^T \mathbf{H} = N\mathbf{I}$, Adachi and Trendafilov (2016) proposed a computationally efficient update of $\mathbf{H}$ and $\mathbf{P}$ as follows: given $\mathbf{P}$, $\hat{\mathbf{H}}$ is the solution to the Orthogonal Procrustes problem; given $\mathbf{H}$, $\hat{\mathbf{P}}$ is updated as follows:

$$\mathbf{P} = T_C\left(\frac{1}{N}\mathbf{Y}^T \mathbf{H}\right). \quad (6)$$

In both (5) and (6), the main complexity involves sorting a matrix of size $J \times Q$, thus, the update can efficiently scale up to high-dimensional setting with very large J .

The AO procedure results in a non-increasing sequence of loss values and converges to a stationary point. However, as (3) is not a convex problem, the algorithm is subjected to local minima. Thus, we implement the algorithm with a multi-start procedure and recommend this option whenever possible: the algorithm is run several times with different starting values and solution with the lowest value is retained as the final solution. The implementation of SEAFa is summarized in Algorithm 1.

Model selection

Algorithm 1 requires six inputs: a matrix of standardized items $\mathbf{Y}$, number of factors Q , number of nonzero loadings C , and two stopping criteria. While all inputs can be user-specified, the number of factors and nonzero loadings may be unknown beforehand. Thus, we propose a sequential data-driven strategy to choose these two inputs:

- Step 1: Number of factors Q can be specified either based on prior knowledge or by using different factor retention methods such as parallel analysis (Horn, 1965), scree plot (Cattell, 1966), Kaiser-Guttman rule (Kaiser, 1960), or a machine learning method (Goretzko & Bühner, 2020). The *seafar* package implements the first three methods.

Algorithm 1: Sparse Exploratory Approximate Factor Analysis (SEAFA)**Input:** $\mathbf{Y}, Q, C, \text{MaxIter}, \epsilon$ **Output:** $\hat{\mathbf{P}}, \hat{\mathbf{H}}$ Initialize loading matrix $\mathbf{P}^{(0)}$ having C nonzero loadings**while** $\Delta \text{lossfunction} > \epsilon$ **or** $\text{iteration} == \text{MaxIter}$ **do** **if** η_q are orthogonal **then** $\mathbf{H} \leftarrow \sqrt{N}\mathbf{U}\mathbf{V}$ from the SVD of $N^{-1/2}\mathbf{Y}\mathbf{P} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ $\mathbf{A} \leftarrow N^{-1}\mathbf{Y}^T\mathbf{H}$ **else** **while** $\Delta \text{lossfunction2} > \epsilon$ **or** $\text{iteration} == \text{MaxIter}$ **do** **for** $q \leftarrow 1$ **to** Q **do** $\mathbf{E}_q \leftarrow \mathbf{Y} - \mathbf{H}\mathbf{P}_q^T + \eta_q \mathbf{p}_q^T$ $\eta_q \leftarrow \frac{\sqrt{N}\mathbf{E}_q \mathbf{p}_q}{\sqrt{(\mathbf{E}_q \mathbf{p}_q)^T (\mathbf{E}_q \mathbf{p}_q)}}$ **end** **end** $\alpha \leftarrow$ maximum eigenvalue of $\mathbf{H}^T \mathbf{H}$ $\mathbf{A} = (\mathbf{P}^T - \frac{\mathbf{H}^T \mathbf{H} \mathbf{P}^T - \mathbf{H}^T \mathbf{Y}}{\alpha N})^T$ **end** **for** $q \leftarrow 1$ **to** Q **do** $\mathbf{p}_q := T_{C_q}(\mathbf{a}_q)$ **end****end****return** $\hat{\mathbf{P}}, \hat{\mathbf{H}}$

- Step 2: Based on the selected number of factors, the number of nonzero loadings C is chosen using the Index of Sparseness (Gajjar et al., 2017; Trendafilov, 2014). That is, the algorithm searches through a sequence of candidate values for C and chooses the one that maximizes the Index of Sparseness (IS), which is defined as follows:

$$IS = PEV_{PCA} \times PEV_{rESEM} \times PS, \quad (7)$$

where PEV_{PCA} , PEV_{SEAFA} and PS are the proportion of explained variance using ordinary PCA, the proportion of explained variance using SEAFA, and the proportion of zero loadings, respectively. The IS value increases with increasing PEV_{PCA} , increasing PEV_{SEAFA} , and the proportion of zero loadings. Thus, IS balances goodness-of-fit with the

complexity of the solution (sparser solutions being less complex). A suitable range of C is $Q \times 3$ (at least three nonzero loading per factor) to $J \times Q$.

Note that here in Step 2, PEV_{SEAF} is obtained by fitting SEAF for each candidate value of C using multiple starting values. This procedure can become computationally intensive when the number of variables J is large, as the candidate range for C increases accordingly. To mitigate this issue, we propose an alternative strategy based on warm starts ([Friedman et al., 2010](#)).

Specifically, for a decreasing sequence of candidate values of C , the loading matrix obtained for a given C is used as the starting value for the subsequent (smaller) value of C , while computing the IS at each step. This can help speed up the computational time substantially; however, the results are more dependent on the initial solution, as subsequent estimates build on the loading matrix obtained in the first iteration.

The seafar package

In this section, we demonstrate the use of our package in R using two empirical datasets. The first analysis focuses on recovering the loading patterns of the Big Five dataset (low-dimensional scenario), while the second analysis explores the factor scores of the Autism Genetic dataset from Nishimura et al. ([2007](#)) (ultra-high-dimensional scenario).

Analysis 1: Recovering loading patterns

Data pre-processing

The Big Five dataset is originally obtained from the *qgraph* package in R ([Epskamp et al., 2012](#)), but can be directly loaded from our package. The dataset contains scores of 500 observations on the 240 items of the personality questionnaire. All functions in *seafar* work with standardized items, thus, each variable is standardized to have a mean of zero and unit variance before the analysis:

```
data("big5", package = "qgraph")
big5_std <- scale(big5)
```


Number of factors

In line with the design of the questionnaire used to collect the data, we assume five factors in correspondence with the five personality traits. Alternatively, one can use data-driven different methods to extract number of factors by using the `factor_retention()` function in the `seafar` package. The results can be inspected by calling `summary()` as below:

```
library(seafar)
Q <- factor_retention(data = big5_std)
summary(Q)
```

Number of factors by parallel analysis is: 18

Number of factors by Kaiser rule is: 12

Number of components by scree test is: 5

Analysis with known cardinality

Given the design of the questionnaire (with each of the items constructed to measure one trait), we first focus on a solution with 240 nonzero loadings and the remaining 960 zero. We perform a SEAFAR analysis using a preset cardinality of 240 with a multi-start procedure:

```
big5_multistart <- seafar_multistart(
  data = big5_std,
  nfactors = 5,
  C = 240
)
```

Note that here, we did not restrict the five factors to be orthogonal. Users can specify `orthogonal = TRUE` to impose orthogonality on the factors. The default number of starting values is 50, which can be modified by option `nstarts =`. Users can also specify the number of nonzero loadings *per* factor instead of across all factors. Since the questionnaire was designed to measure each trait with 48 items, we indicate a cardinality of 48 nonzero loadings per factor by changing the input `C = rep(48, 5)`. Furthermore, users can choose the method to initialize the loading matrix by specifying the option `INIT =` with the following choices:

- `INIT = rational`: the initialization is based on the approach by Ding and He (2004): the SVD-based loading matrix is first computed, which are then used to cluster the variables. The resulting cluster structure is then used as a target for rotation. The final rotated loading matrix is used as the initial loading matrix for SEAFA.
- `INIT = random`: the initial loading matrix have complete random values sampled from a multivariate normal distribution $MVN(\mathbf{0}, \mathbf{I})$.
- `INIT = mixed`: the initial loading matrix is a mixed between the random starts (30%) and starting values based on the Singular Value Decomposition (SVD) of the items (70%), which is the default option.

The results can be summarized using `summary()` function with options as below:

```
summary(big5_multistart, disp = "loadings")
```

The number of nonzero loadings is: 240

The estimated loadings matrix is

	[,1]	[,2]	[,3]	[,4]	[,5]
N1	-0.539	0	0	0	0
E2	0	0	-0.37	0	0
O3	0	0	0	0	-0.618
A4	0.303	0	-0.381	0	0
C5	0	0.211	0	0.345	0
N6	-0.476	0	0	0	0
E7	0.319	0	0	0	0
O8	0	0	0	0	-0.51
A9	0	0	-0.432	0	0
C10	-0.333	0	0	0.349	0

The option `disp` = takes two values:

- `disp = "loadings"` only summarizes the loading matrix.
- `disp = "full"` summarizes both the factor score and loading matrices.

The full factor score and loading matrices can be extracted by indicating

`big5_multistart$scores` and `big5_multistart$loadings`, respectively. This is useful when users wish to use the full factor scores and/or loadings for further analyses, for instance, plot a heat map for the loadings.

The above analyses were done using a multi-start procedure. The package also offers a single-start procedure by using the `seafar()` function as follows:

```
big5_result <- seafar(  
  data = big5_std,  
  nfactors = 5,  
  C = 240,  
  INIT = "svd"  
)
```

Here, the `INIT` = input offers an extra options for starting values in addition to the starting configurations above:

- `INIT = svd`: for starting values based on the Singular Value Decomposition of the items.

The single-procedure is computationally faster than the multi-start one, especially when the items matrix is large. However, it does not guarantee to work well in all scenarios but rather is heavily dependent on the type of input data itself. Users are advised to use it with caution.

Analysis with unknown cardinality

Often in an exploratory setting, the number of nonzero loadings is not known beforehand. Here, users can use the `is.seafar()` function for model selection based on the Index of Sparseness. The function requires the number of factors and based on this, it goes through a sequence of 100

possible values for the cardinality and picks the one that results in the highest Index of Sparseness (Gu et al., 2019).

```
model_selection <- is.seafar(  
  data = big5_std,  
  nfactors = 5,  
  INIT = "rational",  
  nstarts = 10,  
  TOL = 3,  
  THR = .2  
)
```

When the dataset is large, this model selection procedure can be computationally intensive. Thus, users can use the `is.seafar_wst()` function that makes use of warm-starts as implemented by the *glmnet* package (Friedman et al., 2010) to speed up the computation time. This function also allows users to specify the number of candidate values to be tested within this range; here, we chose 200 values.

```
model_selection_wst <- is.seafar_wst(  
  data = big5_std,  
  nfactors = 5,  
  INIT = "rational",  
  card.length = 200  
)  
summary(model_selection_wst)
```

The cardinality selected by the Index of Sparseness is: 247

The PEV with this cardinality is: 0.199

The PEV of the unconstrained model (no zero loadings) is: 0.247

The chosen cardinality can be extracted using `summary(model_selection_wst)`, which can then be used in the similar manner as earlier with the `seafar_multistart()` function (or `seafar()` for a single start).

Analysis 2: Investigating the factor scores

As the *seafar* package was developed to specifically deal with high-dimensional setting, we also demonstrate how it can be used to analyze the Autism Genetic data set (Nishimura et al., 2007). The data is publicly available at <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE7329> and can be loaded with the *seafar* package as follows:

```
library(seafar)
data("autism")
```

Note that the data set loaded from the *seafar* package has already been standardized; otherwise, users need to standardize the raw data set prior to the analyses. This data set contains 43893 genetic marker variables for 27 subjects, in which 14 individuals are in the control group, six are in the FMR1-FM autism group, and seven are in the dup15q autism group. We chose three factors as indicated in the original paper by Nishimura et al. (2007). Due to the size of the data set, we perform a model selection with warm-start to determine the number of nonzero loadings from 200 candidate values:

```
autism_cardinality <- is.seafar_wst(
  data = autism,
  nfactors = 3,
  INIT = "rational",
  orthogonal = TRUE,
  card.length = 200
)
selcard <- autism_cardinality$selcard
summary(autism_cardinality)
```

The cardinality selected by the Index of Sparseness is: 37062

The PEV with this cardinality is: 0.251

The PEV of the unconstrained model (no zero loadings) is: 0.321

Using this value of cardinality (i.e., 3.7062×10^4), we run a SEAFAR analysis with multiple starting values

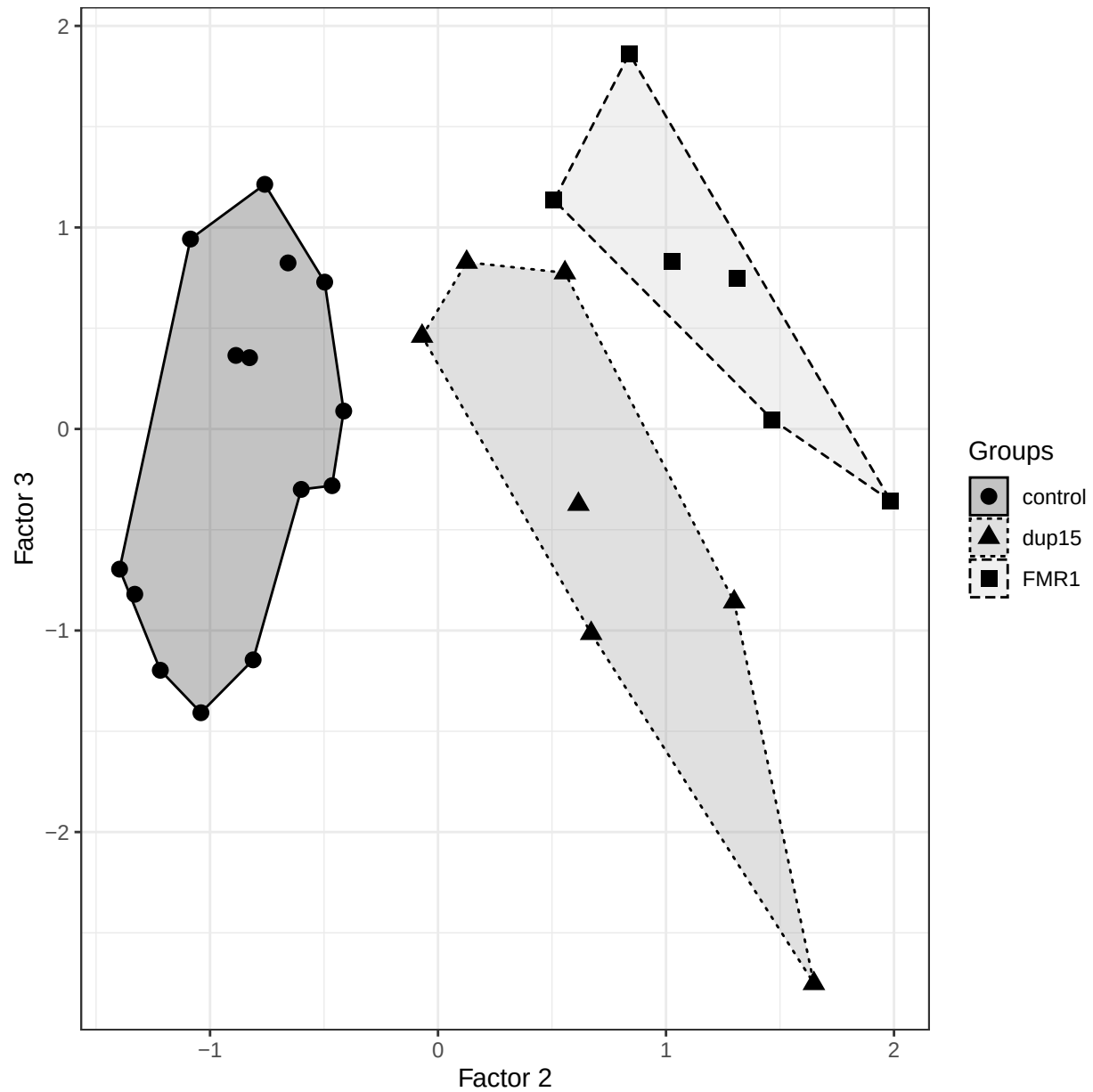
```
autism_multistart <- seafar_multistart(  
  data = autism,  
  nfactors = 3,  
  C = selcard,  
  orthogonal = TRUE,  
  nstarts = 25  
)
```

In this analysis, the factor scores were extracted for subsequent analysis, by using command `autism_multistart$scores`. Figure 1 shows a plot when Factor 2 and Factor 3 were plotted against each other. As can be seen, Factor 2 clearly separated the control group and the two autistic groups. This reflects the same results by Nishimura et al. (2007), without having to use a filtering step (i.e., Nishimura et al. (2007) conducted an ANOVA to select a subset of 293 probes which then be used in a PCA).

Conclusions

In this tutorial, we demonstrated how to perform exploratory factor analysis for complex models and high-dimensional data using the R package *seafar*. The underlying algorithm builds on the approximate factor model and extends the recently proposed Regularized ESEM approach (Le et al., 2026). In particular, *seafar* employs a cardinality constraint to impose simple structure on the loading matrix for both orthogonal and non-orthogonal factors. This constraint offers several advantages. First, it yields exact zero loadings, in contrast to rotation-based methods that rely on post hoc thresholding of small but nonzero values. Second, it provides direct control over the number of nonzero loadings, whereas with the LASSO penalty, for instance, users do not know beforehand how many (non)zero loadings are associated with which level of penalty.

The *seafar* package provides a user-friendly implementation of sparse exploratory factor analysis with minimal input requirements, along with built-in model selection procedures based on the Index of Sparseness. These procedures, suggested by RegularizedSCA (Gu et al., 2019)

Figure 1*Factor scores plot*

and Regularized ESEM (Le et al., 2026), allow users to determine the appropriate number of (non)zero loadings in the absence of strong prior theoretical guidance. The proposed package can be extended to accommodate categorical variables, further broadening its applicability.

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Appendix

Mathematical derivations for SEAFa algorithm

Here, we present the mathematical derivations for the update of factor scores and factor loadings for SEAFa.

1. Update of the factor scores conditional upon fixed values for the loadings is based on the following objective:

$$\begin{aligned} \min_{\eta_{11}, \dots, \eta_{iQ}, \dots, \eta_{NQ}} & \sum_{j=1}^J \sum_{i=1}^N \left(y_{ij} - \sum_q \eta_{iq} p_{jq} \right)^2 + \lambda \sum_{j=1}^J \sum_{q=1}^Q |p_{jq}|_1 \\ \text{subject to} & \sum_i \eta_{iq}^2 = N \quad \forall q = 1, \dots, Q. \end{aligned} \quad (\text{A1})$$

(A1) can be rewritten in matrix form as follows:

$$\begin{aligned} \min_{\boldsymbol{\eta}_1, \dots, \boldsymbol{\eta}_q, \dots, \boldsymbol{\eta}_Q} & \|\mathbf{Y} - \sum_q \boldsymbol{\eta}_q \mathbf{p}_q^T\|^2 + \sum_q \lambda \|\mathbf{p}_q\|_1 \\ \text{subject to} & \boldsymbol{\eta}_q^T \boldsymbol{\eta}_q = N \quad \forall q = 1, \dots, Q. \end{aligned} \quad (\text{A2})$$

Using the method of Lagrange Multipliers, we obtain the Lagrangian function

$$\begin{aligned} \mathcal{L} &= \|\mathbf{Y} - \sum_q \boldsymbol{\eta}_q \mathbf{p}_q^T\|^2 - \mu (\boldsymbol{\eta}_q^T \boldsymbol{\eta}_q - N) \\ &= \|\mathbf{Y} - \sum_{t \neq q} \boldsymbol{\eta}_t \mathbf{p}_t^T - \boldsymbol{\eta}_q \mathbf{p}_q^T\|^2 - \mu (\boldsymbol{\eta}_q^T \boldsymbol{\eta}_q - N) \\ &= \|\mathbf{E}_q - \boldsymbol{\eta}_q \mathbf{p}_q^T\|^2 - \mu (\boldsymbol{\eta}_q^T \boldsymbol{\eta}_q - N) \end{aligned} \quad (\text{A3})$$

Solving $\frac{\partial \mathcal{L}}{\partial \boldsymbol{\eta}_q} = 0$ and $\frac{\partial \mathcal{L}}{\partial \mu} = 0$, we have $\boldsymbol{\eta}_q = \frac{\sqrt{N} \mathbf{E}_q \mathbf{p}_q}{\sqrt{(\mathbf{E}_q \mathbf{p}_q)^T (\mathbf{E}_q \mathbf{p}_q)}}$.

2. The update of factor loadings conditional upon fixed values for the scores is based on the

following objective:

$$\begin{aligned} \min_{\mathbf{P}} & ||\mathbf{Y} - \mathbf{H}\mathbf{P}^T||_2^2 \\ \text{subject to } & \text{Card}(\mathbf{P}) = C, \end{aligned} \tag{A4}$$

which can be rewritten in vectorized form as follows:

$$\begin{aligned} \min_{\mathbf{P}} & ||\text{vec}(\mathbf{Y}) - (\mathbf{I} \otimes \mathbf{H})\text{vec}(\mathbf{P}^T)||_2^2, \\ \text{subject to } & \text{Card}(\mathbf{P}) = C. \end{aligned} \tag{A5}$$

As pointed out by (Adachi & Kiers, 2017), (A5) can be majorized as follows:

$$||\text{vec}(\mathbf{Y}) - (\mathbf{I} \otimes \mathbf{H})\text{vec}(\mathbf{P}^T)||_2^2 \leq c + \alpha ||\mathbf{b} - \text{vec}(\mathbf{P})||_2^2, \tag{A6}$$

where the right-hand side function is also known as the surrogate function. Here, c is a constant with respect to $\mathbf{P}$, α is the maximum eigenvalue of $\mathbf{H}^T \mathbf{H}$, and

$\mathbf{b} = \text{vec}(\mathbf{P}^T - \frac{1}{N\alpha}(\mathbf{H}^T \mathbf{H} \mathbf{P}^T - \mathbf{H}^T \mathbf{Y}))$. The surrogate function in (A6) is minimized when $\text{vec}(\mathbf{P}) = T_C(\mathbf{b})$, where $T_C(\cdot)$ denotes the vector whose C elements with the smallest absolute values are put to zeros.